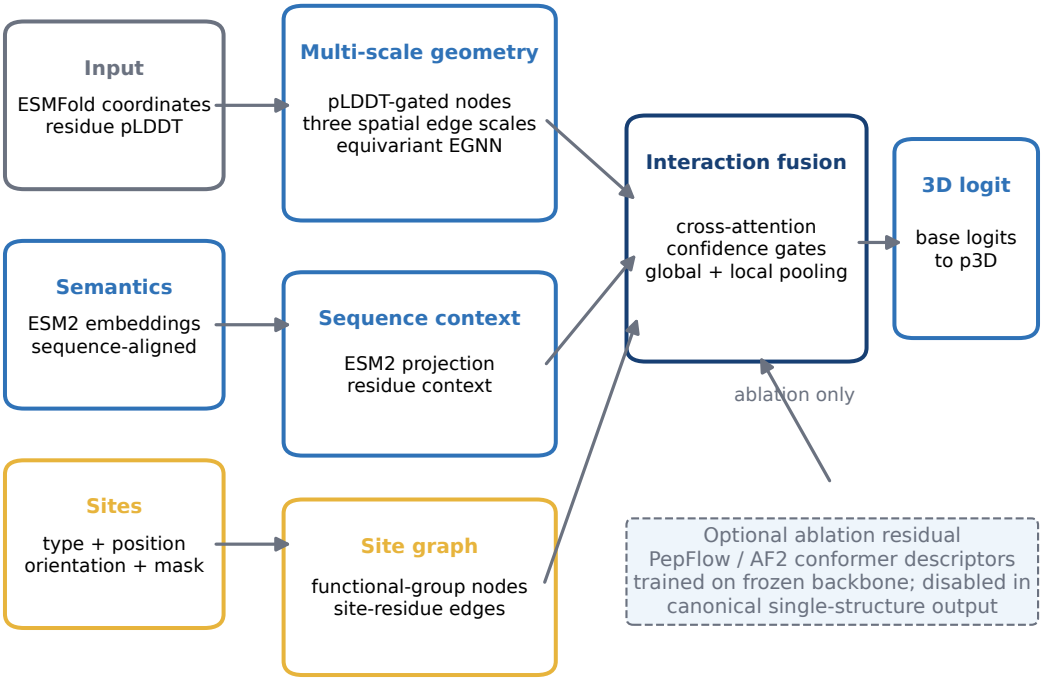
pLDDT-aware multi-scale structural expert
Input
ESMFold coordinates
residue pLDDT
Multi-scale geometry
pLDDT-gated nodes
three spatial edge scales
equivariant EGNN
Semantics
ESM2 embeddings
sequence-aligned
Sequence context
ESM2 projection
residue context
Sites
type + position
orientation + mask
Site graph
functional-group nodes
site-residue edges
Interaction fusion
cross-attention
confidence gates
global + local pooling
3D logit
base logits
to p3D
ablation only
Optional ablation residual
PepFlow / AF2 conformer descriptors
trained on frozen backbone; disabled in
canonical single-structure output